Smart Body Composition System

User Manual

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Author	Nathanael Anak Chulif @ Douglas Chulip
Institution	Universiti Malaysia Sarawak (UNIMAS)

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1. About the System

The Smart Body Composition System is a web-based health tracking platform designed to help you monitor, analyse, and gradually improve your body composition over time. It was developed as a Final Year Project at Universiti Malaysia Sarawak (UNIMAS), with one central idea in mind: your smart scale already collects valuable data — this system helps you do something meaningful with it.

What It Does

At its core, the system is a personal health tracker built around body composition rather than weight alone. Here is what it offers:

- Log body composition measurements from your smart scale — weight, body fat %, muscle mass, visceral fat, bone mass, and more.
- Visualize trends over time with interactive charts covering 7, 14, 30, or 90-day periods, so you can see the bigger picture rather than obsessing over day-to-day fluctuations.
- Receive AI-driven health recommendations personalized to your measurement history, age, and gender — not just generic fitness advice.
- Set and track fitness goals for weight, body fat, muscle mass, and visceral fat, each with a target value and a deadline.
- Stay on track with weekly reports and measurement reminders delivered through the in-app notification system.

Who It Is For

This system was built with a broad audience in mind. If you own a body composition smart scale — whether it's a Tanita, Omron, InBody, Xiaomi, or any similar device — and you want to do more than just glance at the numbers, this is for you. It is equally useful for:

- Fitness enthusiasts wanting to go beyond weight and track body fat %, muscle gain, and metabolic health over time.
- Anyone who has ever wondered what their body composition readings actually mean and how they are changing.

Key Features at a Glance

Feature	Description
10 Body Metrics	Weight, body fat %, body fat kg, muscle mass, bone mass, body water %, visceral fat, calories (kcal), body age, physical rating
Trend Charts	Interactive line graphs for any metric over 7/14/30/90 days
AI Recommendations	Personalised health insights generated from your data
Goal Tracking	Set targets with deadlines and monitor progress
Notifications	Weekly reports, measurement reminders, goal-achievement alerts
Admin Panel	Manage users, system settings, and recommendation templates

Unit Conversion	Toggle between metric (kg/cm) and imperial (lb/in)

⚠ **Important:** This system does not connect directly to your smart scale. You will need to manually enter the readings shown on your scale's display into the system. This is by design — it keeps the system compatible with any brand of smart scale, regardless of whether it supports Bluetooth or app integration.

2. System Requirements

Most users will access the system through a web browser and need nothing more than a stable internet connection. If you are setting up the system locally — perhaps as a developer or system administrator — there are a few more prerequisites to take care of first.

For Users (Accessing the Deployed System)

- A modern web browser — Google Chrome, Mozilla Firefox, Microsoft Edge, or Opera (latest versions recommended).
- An internet connection.

That is genuinely all you need. The system runs entirely in the browser with no downloads or plugins required.

For Local Installation (Developer / Administrator)

If you are running the system on your own machine, make sure the following software is installed before you begin:

Software	Minimum Version	Purpose
XAMPP	8.2+	Local Apache + MySQL + PHP server
PHP	8.2 or higher	Server-side runtime
Composer	2.x	PHP dependency manager
Node.js	20.x or higher	Frontend build tools (includes npm)
MySQL / MariaDB	5.7 or higher	Database (included with XAMPP)
Git	Any version	For cloning the repository

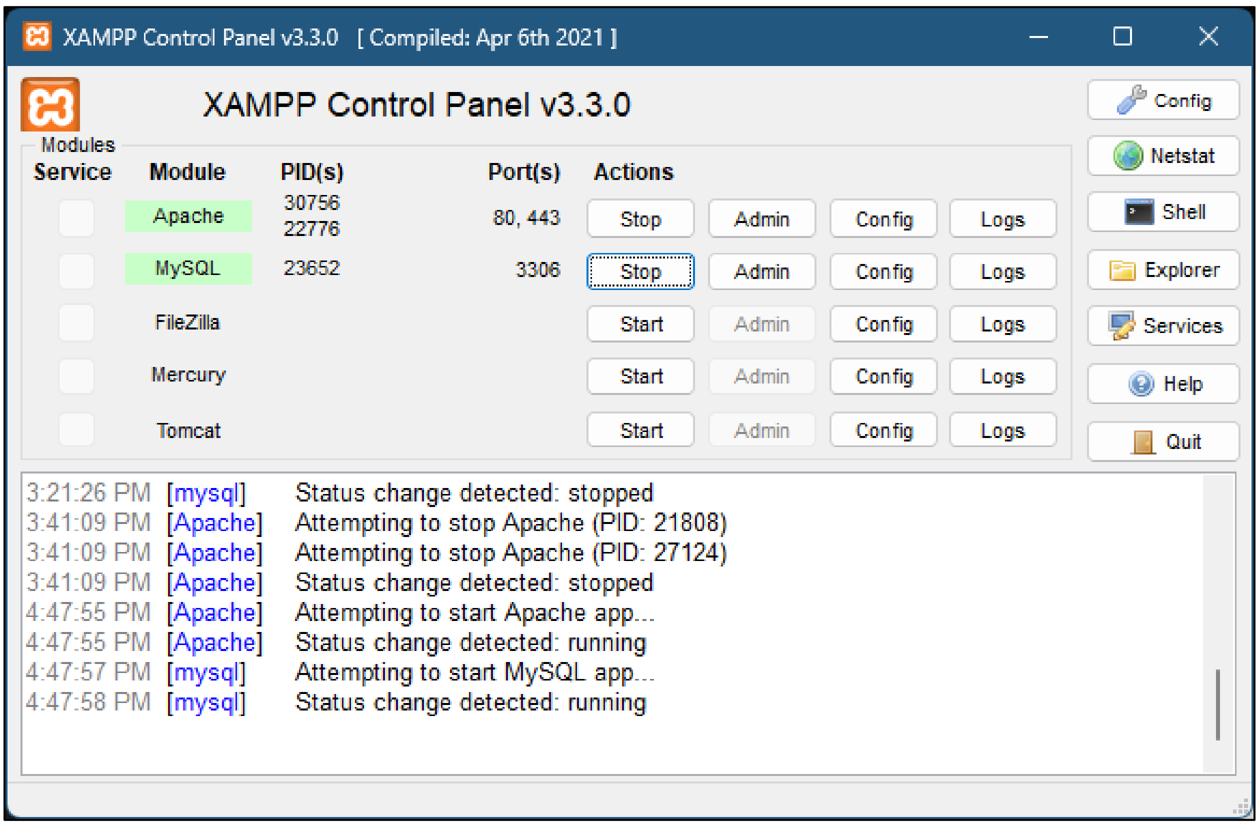
Note: XAMPP is the recommended choice for Windows users because it bundles Apache, MySQL, and PHP into a single installer — no configuration headaches. Download it from apachefriends.org.

3. Installation Guide

What follows is a step-by-step walkthrough for getting the system running on your local machine. If you are accessing a hosted version of the system — for example, deployed on Render.com — you can skip ahead to Section 4: First-Time Setup. For everyone else, follow the steps below in order.

Step 1 — Install XAMPP

1. Download XAMPP from https://www.apachefriends.org.
2. Run the installer and accept the default settings — the defaults work perfectly for this project.
3. Once the installation completes, open the XAMPP Control Panel.
4. Click Start next to both Apache and MySQL.
5. Both services should show a green background with "Running" status. If either fails to start, see the Troubleshooting section.



Step 2 — Install Composer

6. Download the Composer installer from https://getcomposer.org.
7. Run the installer. When prompted, point it to the PHP executable inside your XAMPP folder — for most Windows users, this will be at C:\xampp\php\php.exe.
8. Once the installation finishes, verify that it worked by opening PowerShell or Command Prompt and running:

```
composer --version
```

Step 3 — Install Node.js

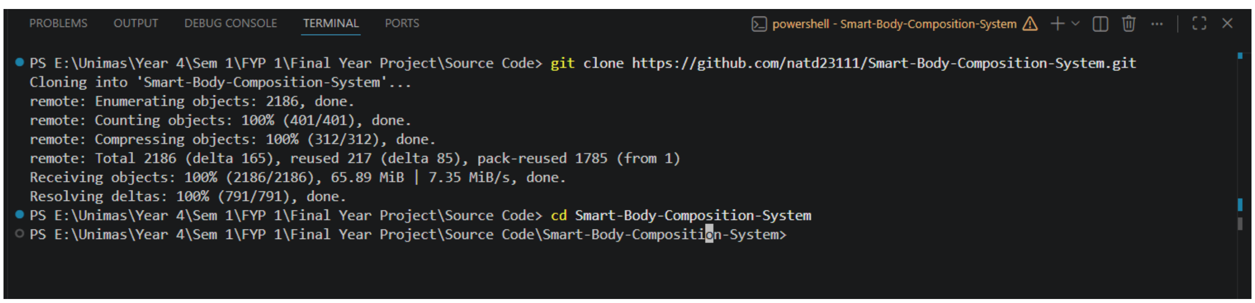
9. Download the LTS version of Node.js from https://nodejs.org. The LTS release is the most stable.
10. Run the installer with the default settings.
11. Verify both Node.js and npm installed correctly:

```
node --version
npm --version
```

Step 4 — Clone the Repository

With the prerequisites in place, it is time to download the project itself. Open PowerShell or Command Prompt and run:

```
git clone https://github.com/natd23111/Smart-Body-Composition-System.git
cd Smart-Body-Composition-System
```



Step 5 — Install PHP Dependencies

Inside the project folder, run Composer to download all required PHP packages:

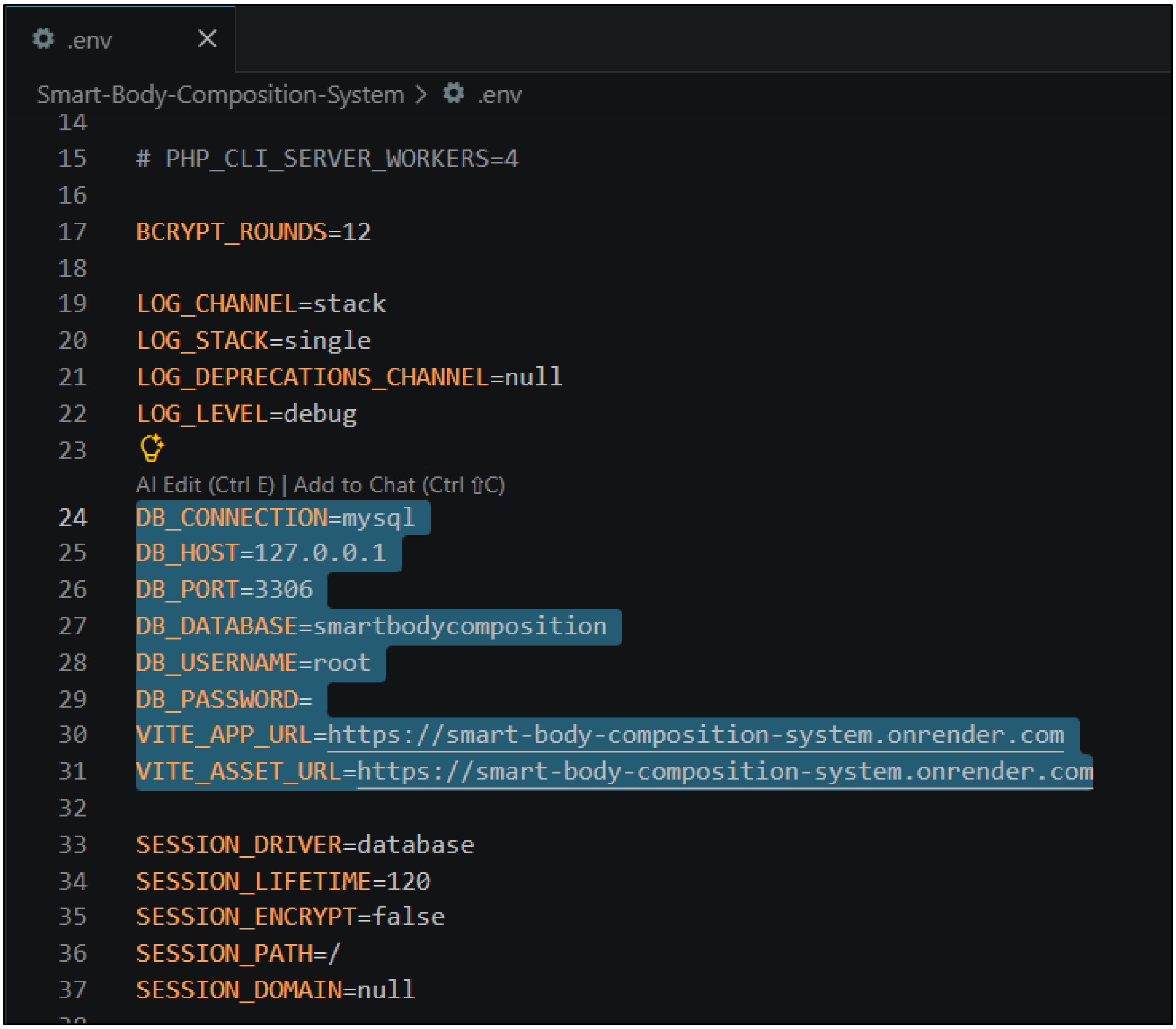
```
composer install
```

Depending on your internet speed, this may take anywhere from two to five minutes. You will see packages downloading one by one — that is completely normal.

Step 6 — Configure Environment Variables

The application reads its configuration — database credentials, app keys, mail settings — from a file called .env. Create yours by copying the provided example:

```
copy .env.example .env
```



Open the new .env file in any text editor (Notepad, VS Code, and Notepad++ all work fine) and update the database section:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=smartbodycomposition
DB_USERNAME=root
DB_PASSWORD=              # Leave empty for XAMPP default
```

Step 7 — Generate Application Key

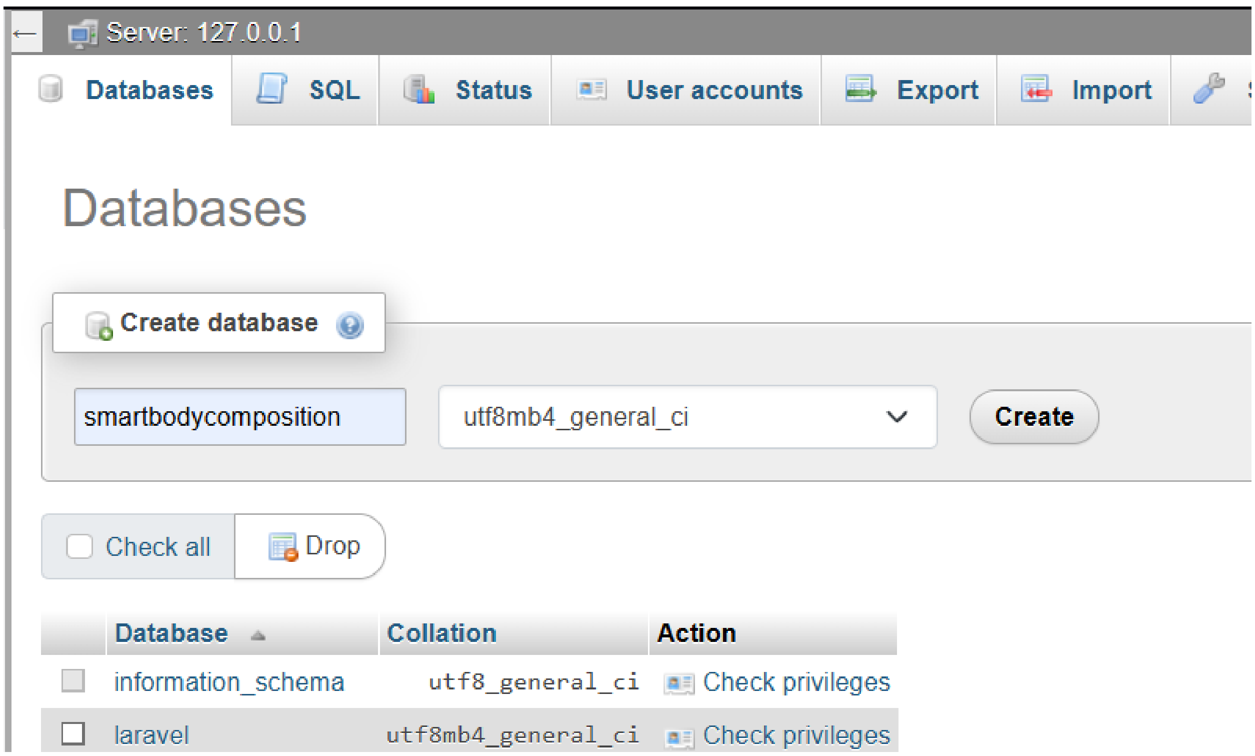
Laravel requires a unique application key for encrypting session data and cookies. Generate one with:

```
php artisan key:generate
```

You should see the confirmation message: Application key set successfully.

Step 8 — Create Database & Run Migrations

12. Open your browser and navigate to http://localhost/phpmyadmin.
13. Click New on the left sidebar.
14. Enter smartbodycomposition as the database name, then click Create.



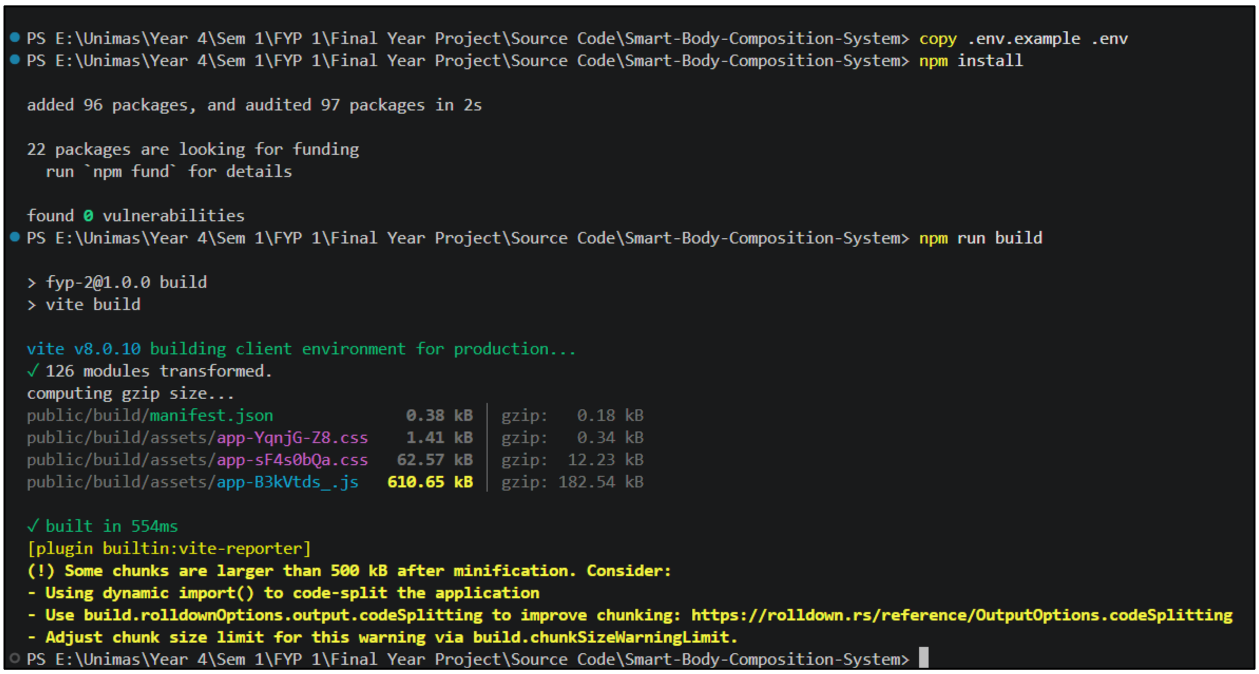
15. Back in the terminal, run the migrations to create all the necessary tables and populate them with demo data:

```
php artisan migrate --seed
```

Step 9 — Build Frontend Assets

The frontend is built using Vite and needs to be compiled before the application can serve its pages correctly:

```
npm install
npm run build
```



```
PS E:\Unimas\Year 4\Sem 1\FYP 1\Final Year Project\Source Code\Smart-Body-Composition-System> copy .env.example .env
PS E:\Unimas\Year 4\Sem 1\FYP 1\Final Year Project\Source Code\Smart-Body-Composition-System> npm install

added 96 packages, and audited 97 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS E:\Unimas\Year 4\Sem 1\FYP 1\Final Year Project\Source Code\Smart-Body-Composition-System> npm run build

> fyp-2@1.0.0 build
> vite build

vite v8.0.10 building client environment for production...
✓ 126 modules transformed.
computing gzip size...
public/build/manifest.json              0.38 kB │ gzip:   0.18 kB
public/build/assets/app-YqnjG-Z8.css    1.41 kB │ gzip:   0.34 kB
public/build/assets/app-sF4s0bQa.css   62.57 kB │ gzip:  12.23 kB
public/build/assets/app-B3kVtds_.js   610.65 kB │ gzip: 182.54 kB

✓ built in 554ms
[plugin builtin:vite-reporter]
(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rolldownOptions.output.codeSplitting to improve chunking: https://rolldown.rs/reference/OutputOptions.codeSplitting
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
PS E:\Unimas\Year 4\Sem 1\FYP 1\Final Year Project\Source Code\Smart-Body-Composition-System>
```

Step 10 — Composer Setup

```
composer setup
```

This single command automates dependency installation, environment setup, key generation, database migrations, seeding, and the frontend build. You still need to edit your .env file with the correct database credentials before running it.

Step 11: Start Server

Everything is in place. Start the development server with:

```
php artisan serve
```

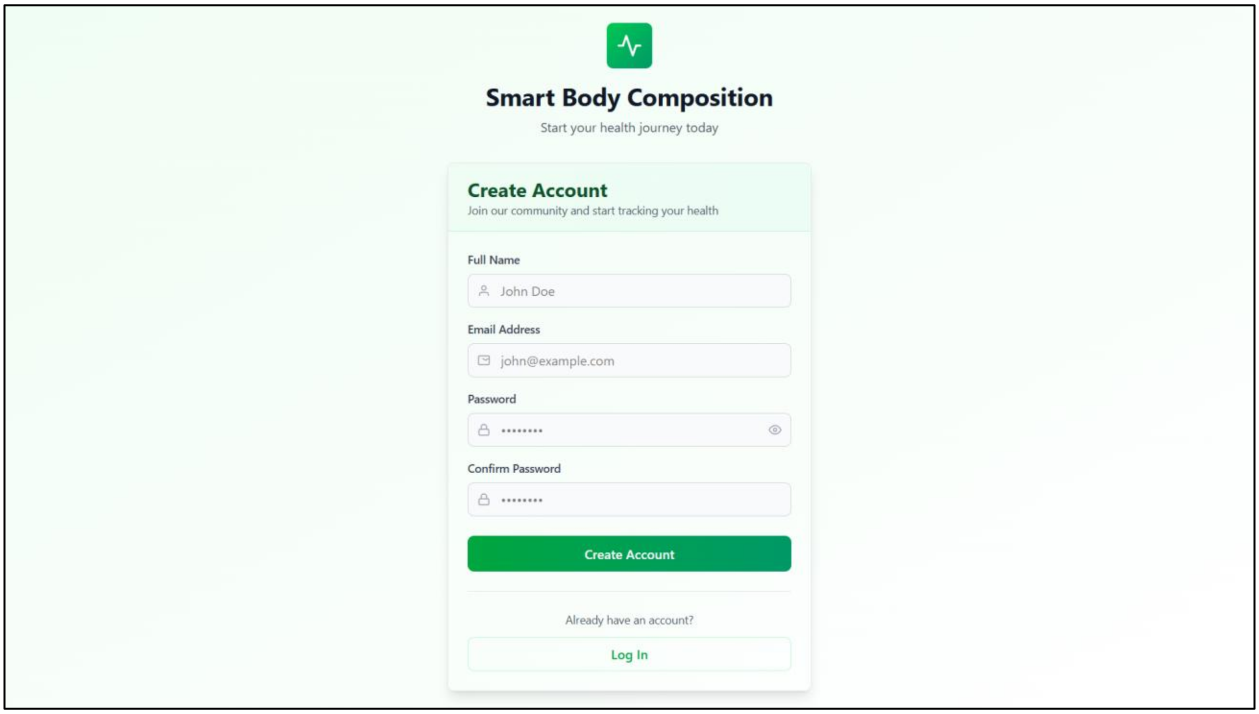
Then open your browser and visit http://127.0.0.1:8000. If you see the login page, the installation was successful — proceed to Section 4 to create your account.

4. First-Time Setup

Once you can reach the login page — whether through a local installation or a hosted deployment — the next step is to create your account and provide the profile information that allows the system to personalize its recommendations for you.

Creating an Account

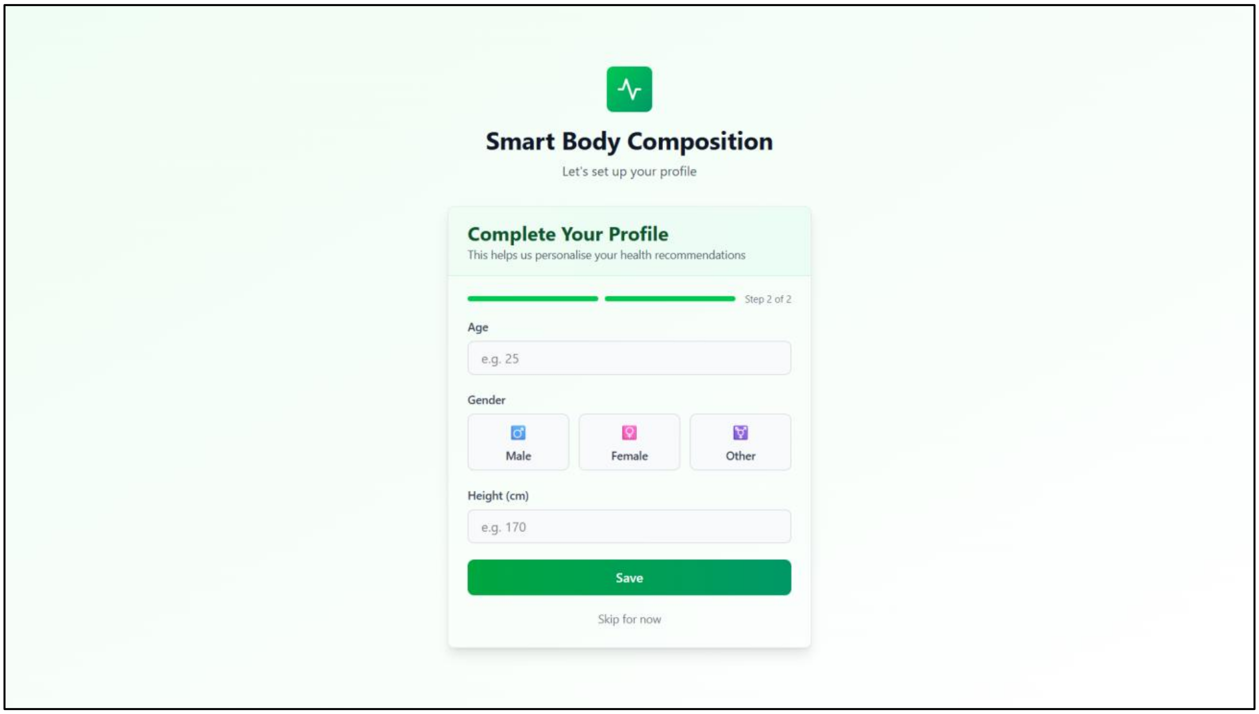
16. On the login page, click the Sign up link.



17. Fill in the registration form with your details:
    - Name — your full name.
    - Email — a valid email address you have access to.
    - Password — at least 8 characters.
    - Confirm Password — re-enter the same password to confirm.
18. Click Create Account. You will be redirected automatically to the profile setup page.

Setting Up Your Profile

This is an important step that is easy to rush through — please take a moment to fill it in accurately. The system uses your age, gender, and height to calibrate its health recommendations; the reference ranges for body fat %, muscle mass, and other metrics differ significantly between a 22-year-old man and a 45-year-old woman, for instance.



Field	Description	Example
Age	Your current age in years	25
Gender	Select Male or Female	Male
Height	Your height in centimetres (or inches, if using imperial units)	170

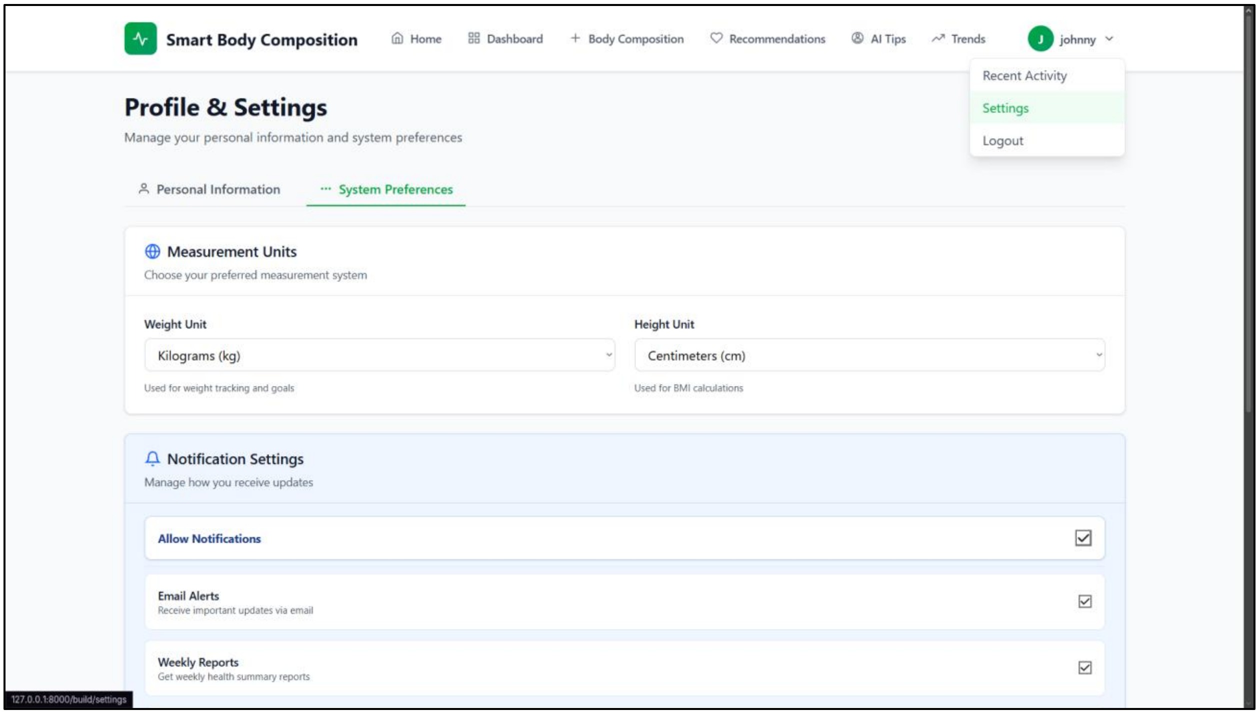
Note: The system uses your age, gender, and height to personalise health recommendations. Reference ranges for body fat %, muscle mass, and other metrics vary by these factors. Providing accurate data ensures your recommendations are relevant.

19. Click Save to continue.

Choosing Your Unit System

Head over to Settings to choose the unit system you prefer:

- Metric — weight in kilograms (kg), height in centimetres (cm).
- Imperial — weight in pounds (lb), height in inches (in).



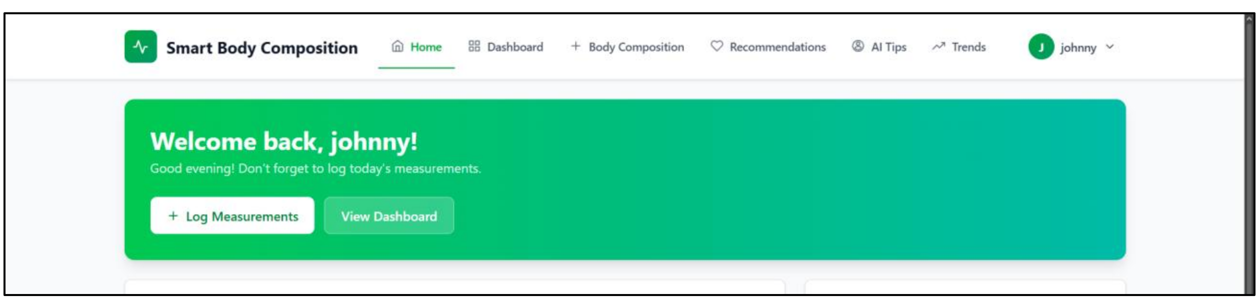
Whichever you choose, all measurements you enter going forward will follow that unit system consistently throughout the application.

5. User Interface Overview

Before diving into specific features, it helps to get a feel for how the interface is organized. The layout is intentionally simple: on the top navigation.

Top bar Navigation

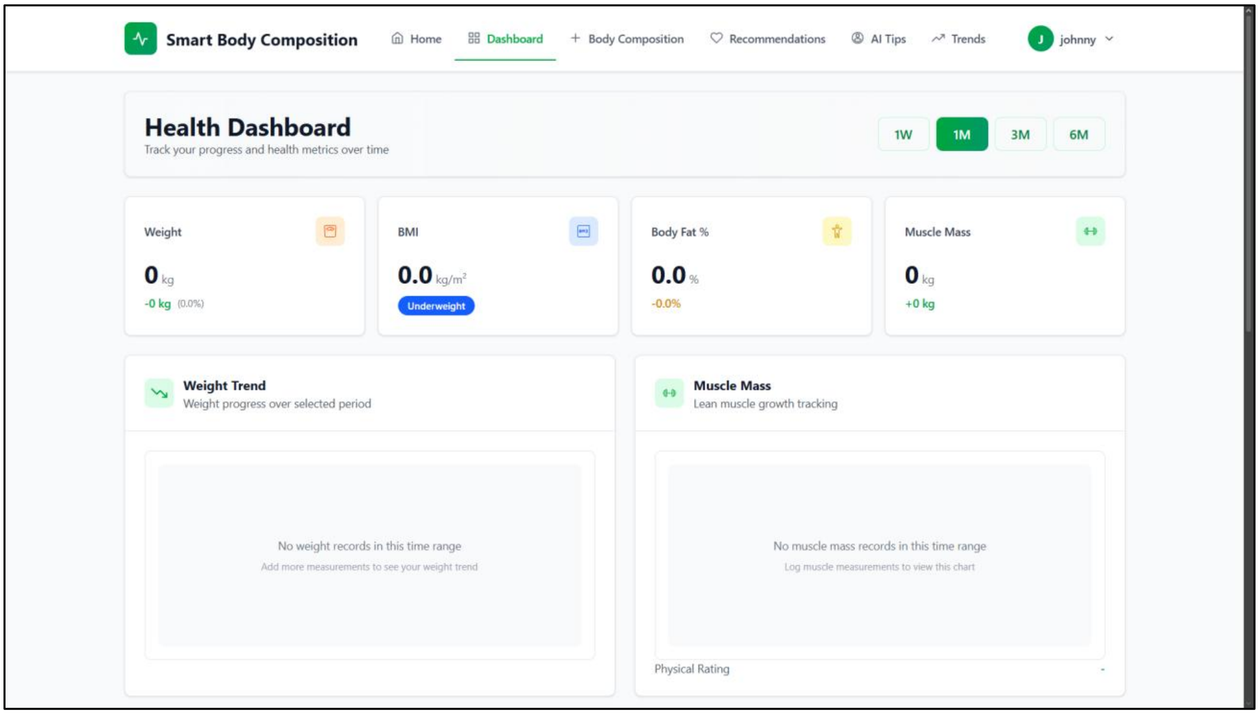
Everything in the system is reachable from the top bar. Here is a quick reference for what each item does:



Menu Item	Purpose
Home	Overview of your recent activity, quick actions, and goals.

Dashboard	Overview of your health metrics, latest measurements, and recent trends
Body Composition	Add, view, edit, and delete body composition records
Recommendations	View AI-generated health insights
AI Tips	Quick health tips based on your profile
Trends	Visualize measurement changes over time

Dashboard Layout



The Dashboard is your home base — the first page you see after logging in. It is designed to give you a complete snapshot at a glance without needing to navigate anywhere:

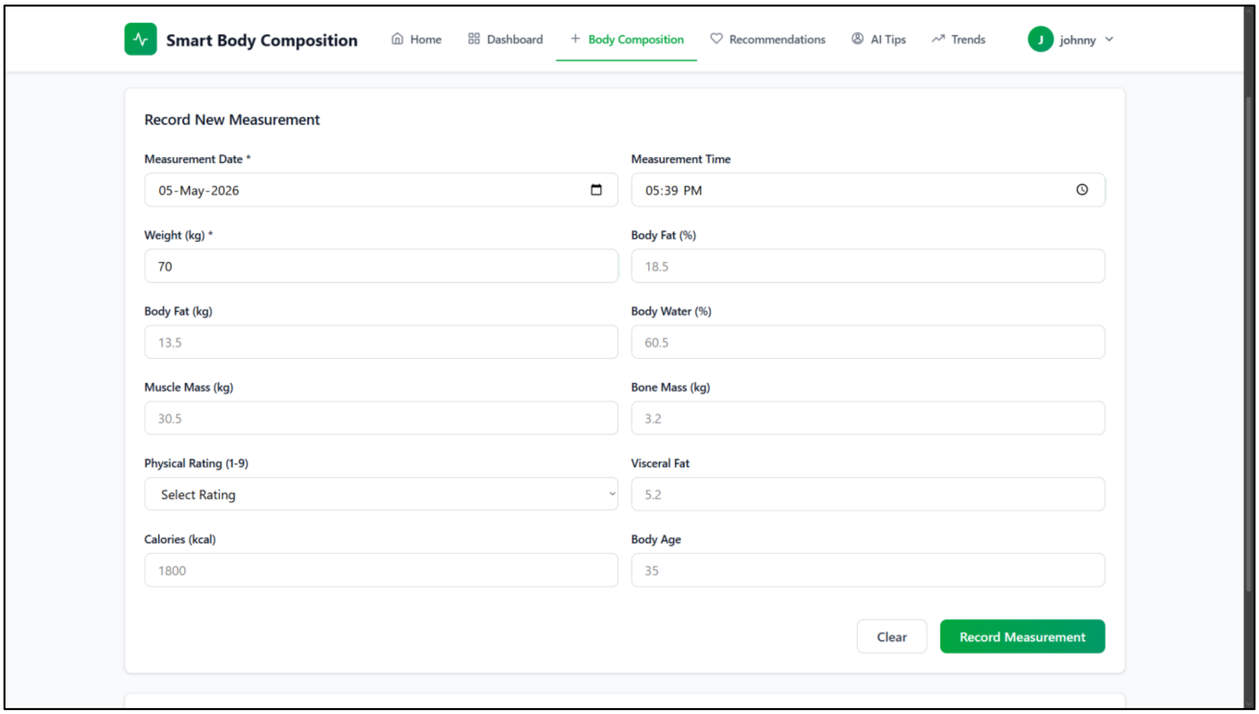
- Summary cards — your latest values for key metrics (weight, body fat, muscle mass, and more).
- Latest measurement — your most recent full body composition record in one place.
- Active goals — a quick view of the goals you are currently working toward and your current progress against each.
- Recent trends — compact mini-charts showing how your key metrics have moved recently.

6. Entering Body Composition Data

Recording your measurements is the most fundamental action in the system — everything else, from the trend charts to the AI recommendations, depends on the data you enter here. The process is straightforward: after stepping off your smart scale, open this page and key in exactly what the scale shows.

Step-by-Step

20. Click Body Composition in the sidebar.
21. Click the Add New Record button at the top of the page.

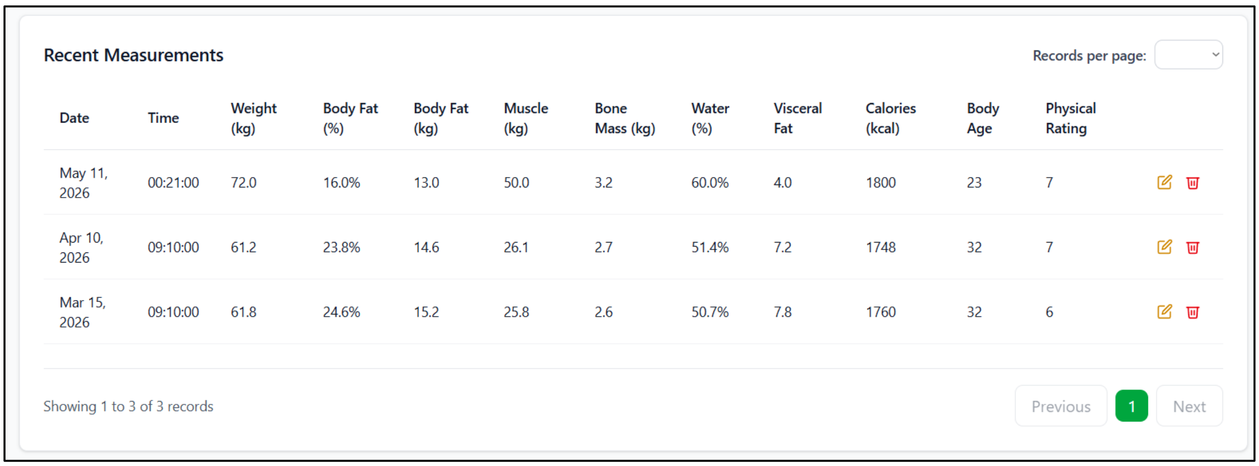


22. Fill in each field with the reading shown on your scale (refer to the table below if you are unsure what any metric means).
23. Click Save Record.

Understanding Each Metric

#	Metric	Description	Typical Healthy Range
1	Weight (kg)	Your total body weight	Varies by height/build
2	Body Fat (%)	Percentage of your body that is fat tissue	Men: 10–20%, Women: 18–28%
3	Body Fat (kg)	Absolute weight of fat tissue	Calculated from weight × body fat %
4	Muscle Mass (kg)	Weight of skeletal muscle	Men: ~35–45 kg, Women: ~25–35 kg

5	Bone Mass (kg)	Weight of bone mineral content	2.5–4.0 kg typical
6	Body Water (%)	Hydration level — percentage of body that is water	45–65%
7	Visceral Fat	Fat around internal organs (scale 1–59)	1–12 is healthy
8	Calories (kcal)	Your Basal Metabolic Rate (BMR) — calories burned at rest	1200–2500 typical
9	Body Age	Metabolic age estimate based on your composition	Should be close to actual age
10	Physical Rating	Overall fitness classification (scale 1–9)	5–9 is active/athletic


Recent Measurements

Records per page:

Date	Time	Weight (kg)	Body Fat (%)	Body Fat (kg)	Muscle (kg)	Bone Mass (kg)	Water (%)	Visceral Fat	Calories (kcal)	Body Age	Physical Rating
May 11, 2026	00:21:00	72.0	16.0%	13.0	50.0	3.2	60.0%	4.0	1800	23	7
Apr 10, 2026	09:10:00	61.2	23.8%	14.6	26.1	2.7	51.4%	7.2	1748	32	7
Mar 15, 2026	09:10:00	61.8	24.6%	15.2	25.8	2.6	50.7%	7.8	1760	32	6

Showing 1 to 3 of 3 records

Previous 1 Next

Tips for Accurate Measurements

The quality of your trend data depends heavily on how consistently you take measurements. A few habits make a significant difference:

- Measure at the same time every day — ideally first thing in the morning, after using the bathroom and before eating or drinking anything.
- Stand barefoot on the scale with both feet squarely on the metal electrodes. Socks or shoes will interfere with the bioelectrical impedance reading.
- Keep the scale on a hard, flat surface. Carpet absorbs and distorts the impedance signal.
- Enter the date and time of each measurement accurately — the system uses these timestamps for all trend calculations.
- Aim for at least one measurement per week. More frequent entries produce smoother, more meaningful trend lines.

Managing Records

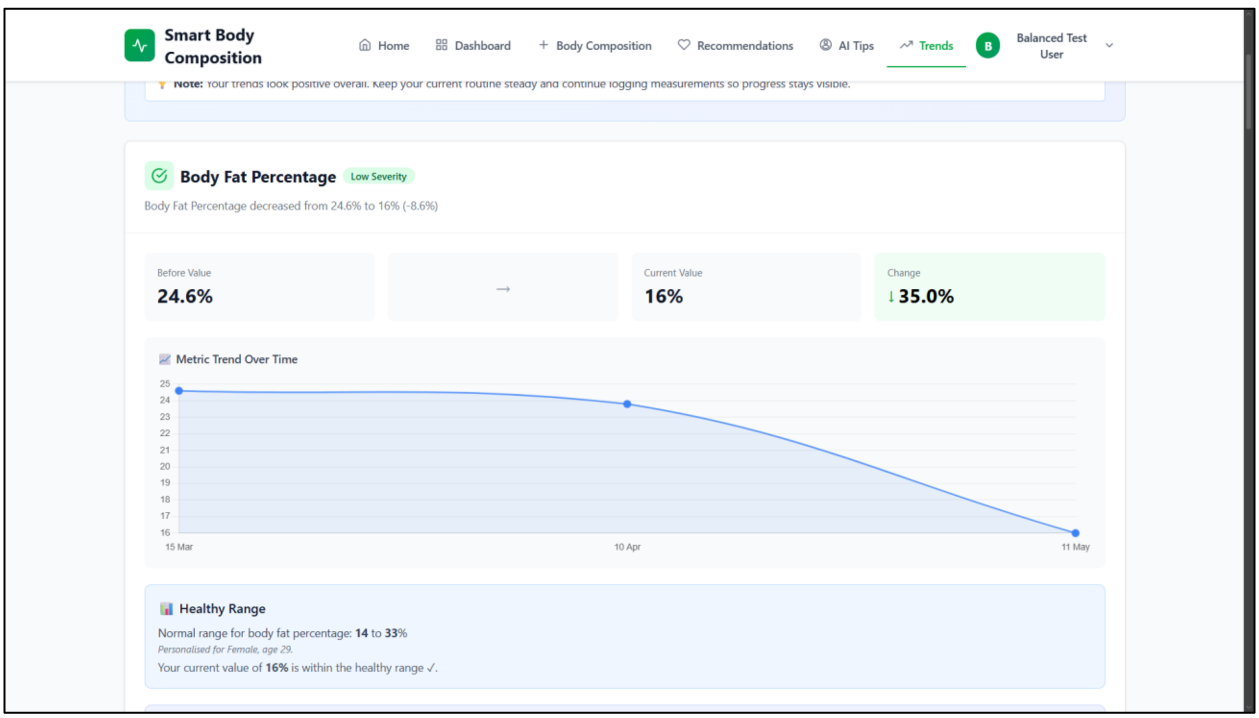
All your records are listed in a table on the Body Composition page. You can edit a record by clicking the edit icon next to it — useful if you mistyped a value — or delete it entirely by clicking the delete icon. The system will ask you to confirm before permanently removing any record.

7. Viewing Trends

A single measurement tells you where you are today. A trend tells you where you are heading — which is far more useful. The Trends page visualises your body composition changes over time as interactive line charts, making it easy to spot patterns that would be invisible from isolated readings.

How to Use

24. Click Trends in the sidebar.
25. Use the time period selector to choose your window:
    - 7 days — a one-week snapshot, useful for spotting short-term fluctuations.
    - 14 days — two weeks of context.
    - 30 days — a full month, generally the most useful starting point for identifying real trends.
    - 90 days — three months, ideal for seeing sustained progress or long-term drift.



26. The chart updates automatically as soon as you change the period. Hover over any data point to see the exact value and date.

Reading the Charts

- X-axis (horizontal) — the dates of your measurements.
- Y-axis (vertical) — the value of the metric being displayed (weight, fat %, etc.).
- Connected dots — each dot is one measurement entry; the line connecting them shows the direction of change.
- Gaps in the line — periods where no measurement was recorded. This is a useful visual reminder to stay consistent.

What to Look For

Rather than chasing a specific number on any given day, focus on the direction and shape of the trend:

Goal	What a good chart looks like
Losing weight	A gradual downward slope (not a sharp drop)
Gaining muscle	A gradual upward slope on the muscle mass chart
Reducing body fat	A steady decline in body fat % and body fat kg
Healthy hydration	Body water % staying consistently between 45–65%
Lowering visceral fat	A declining number, ideally below 12

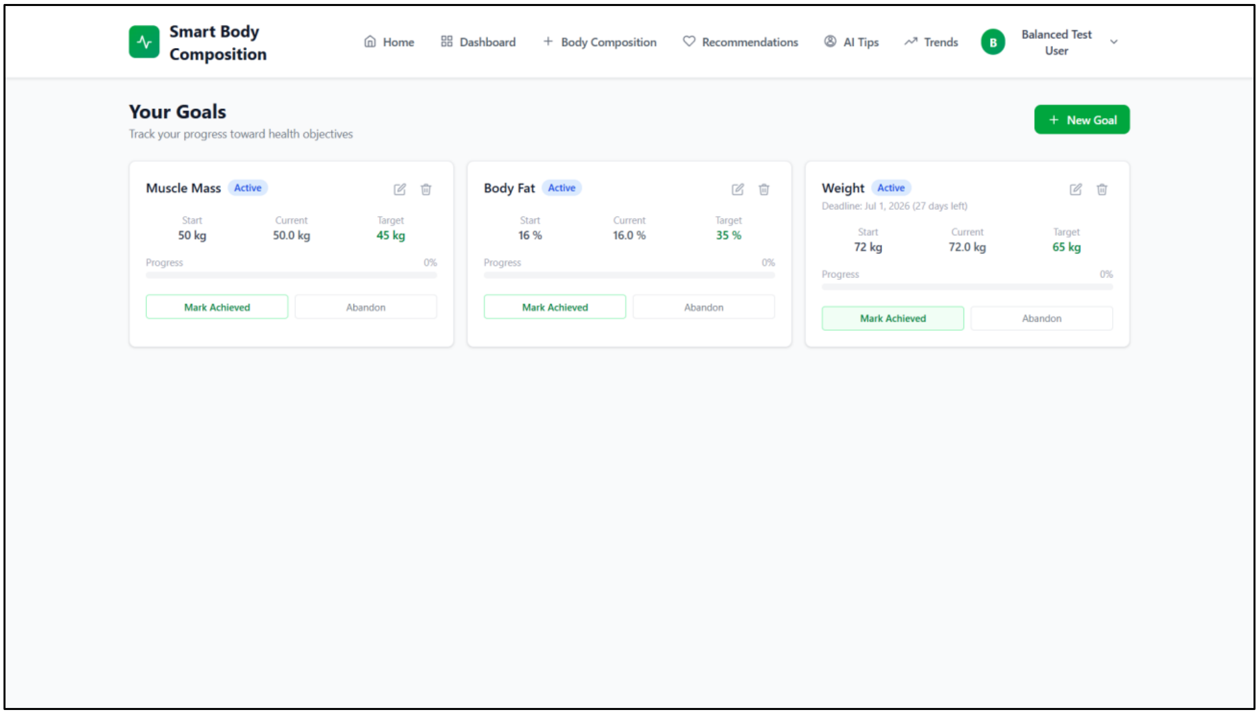
Note: Healthy change is slow and steady. A drop of more than 1 kg per week typically reflects water loss rather than actual fat reduction — and it is not sustainable. Patience matters more than speed here.

8. Setting & Tracking Goals

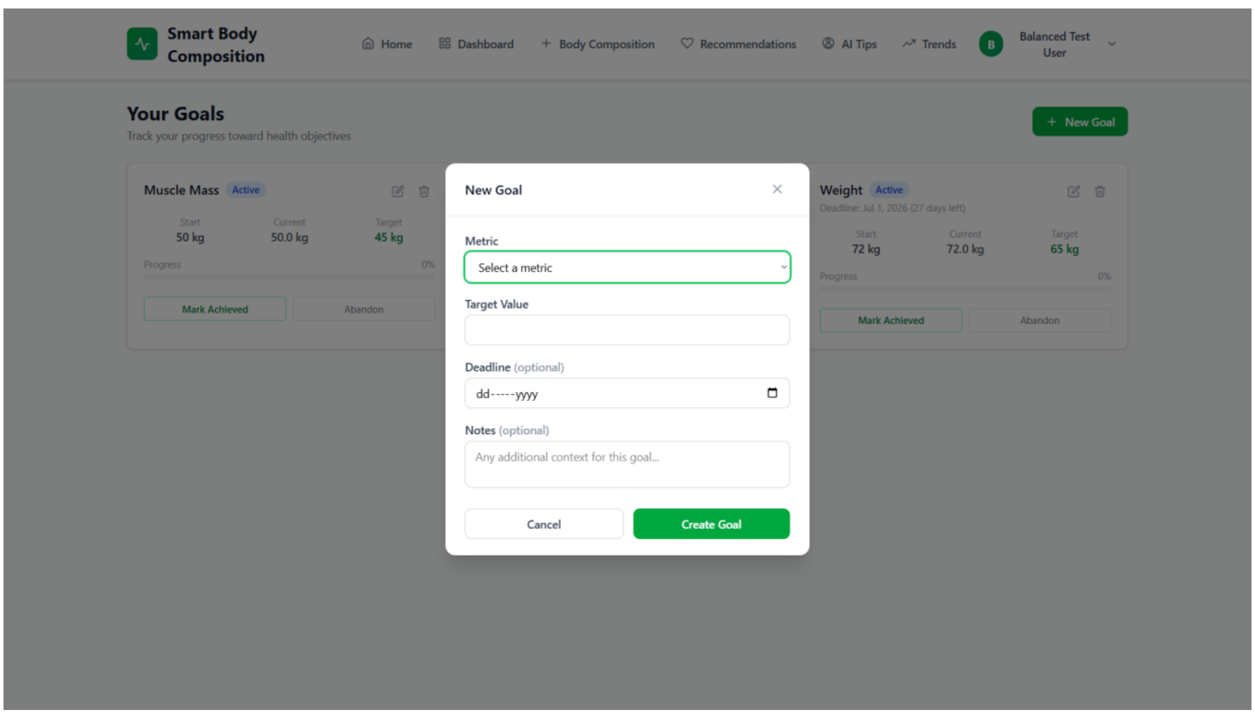
Tracking data is motivating on its own, but having a concrete target to work toward makes a real difference. The Goals feature lets you define exactly what you want to achieve, when you want to achieve it, and then automatically tracks your progress each time you log a new measurement.

Creating a Goal

27. Click My Goals in the sidebar.



28. Click the New Goal button.



29. Fill in the goal form:

Field	Description	Example

Metric	What you want to improve	Weight (kg), Body Fat (%), Muscle Mass (kg), or Visceral Fat
Target Value	Your desired measurement	78.0 (if you want to weigh 78 kg)
Start Value	Your current measurement (auto-filled from latest record)	88.2
Deadline	Date by which you want to achieve the target	2026-07-31
Notes	Any additional notes about your strategy	"Caloric deficit + 3x gym per week"

30. Click Create Goal.

Monitoring Progress

Once a goal is saved, the system takes over the tracking. Each goal card displays a progress bar that updates automatically whenever you log a new measurement — there is nothing else you need to do manually. Goals sit in one of three states:

- Active — you are still working toward the target.
- Achieved — congratulations: you have hit your target value.
- Abandoned — you have chosen to stop tracking this particular goal.

Editing or Deleting Goals

Life changes, and so do targets. If you need to adjust a goal's deadline, update a target value, or add new notes about your approach, click the edit icon on the goal card. To remove a goal entirely, click the delete icon — but note that this action is permanent.

9. Health Recommendations

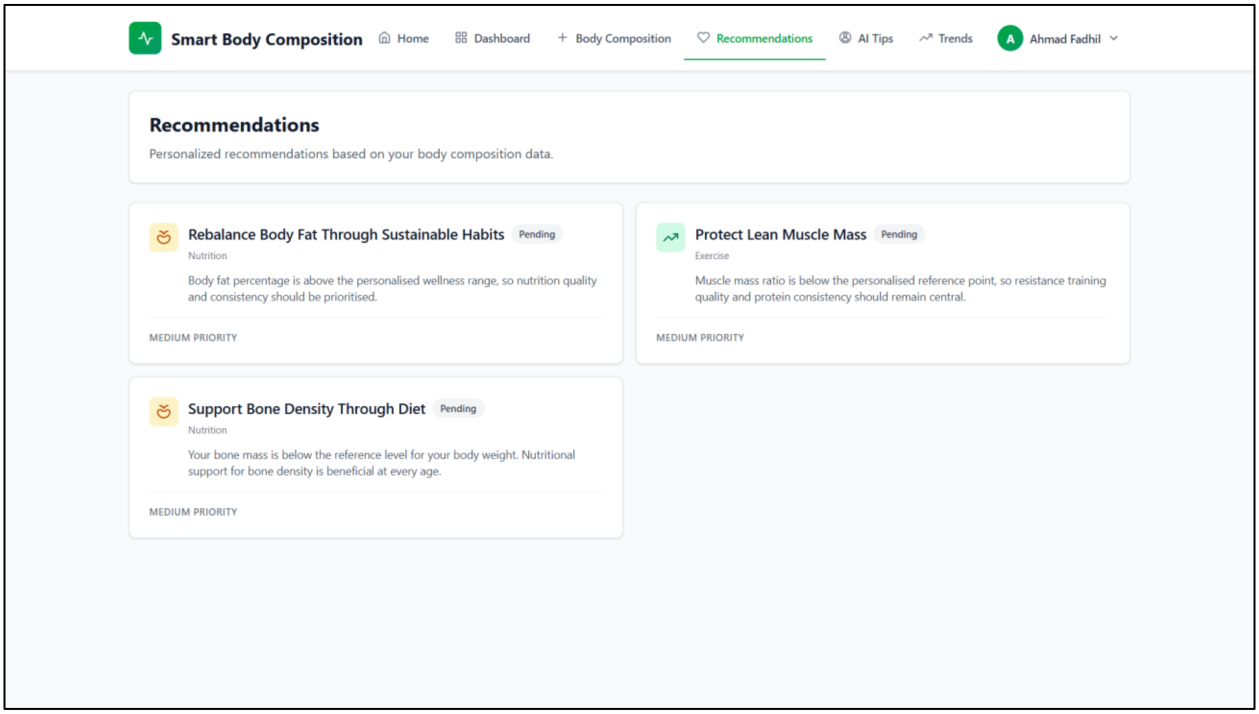
One of the more distinctive features of this system is its ability to generate health recommendations that are tailored to your specific data, rather than serving up the same generic advice to every user. The recommendations engine looks at how your metrics have changed and compares them against reference ranges calibrated to your age and gender.

How Recommendations Work

When you trigger a recommendation generation, the system compares your earliest measurement to your most recent one within the selected period. It then evaluates each metric against appropriate reference ranges — adjusted for your age and gender — and produces an advice card for any metric that falls outside the healthy range or shows a concerning trend. If everything looks good, it says so.

Generating Recommendations

31. Click Recommendations in the sidebar.
32. Make sure you have at least two measurements recorded — the engine needs a before and after to detect any kind of trend.
33. Click the Generate button.



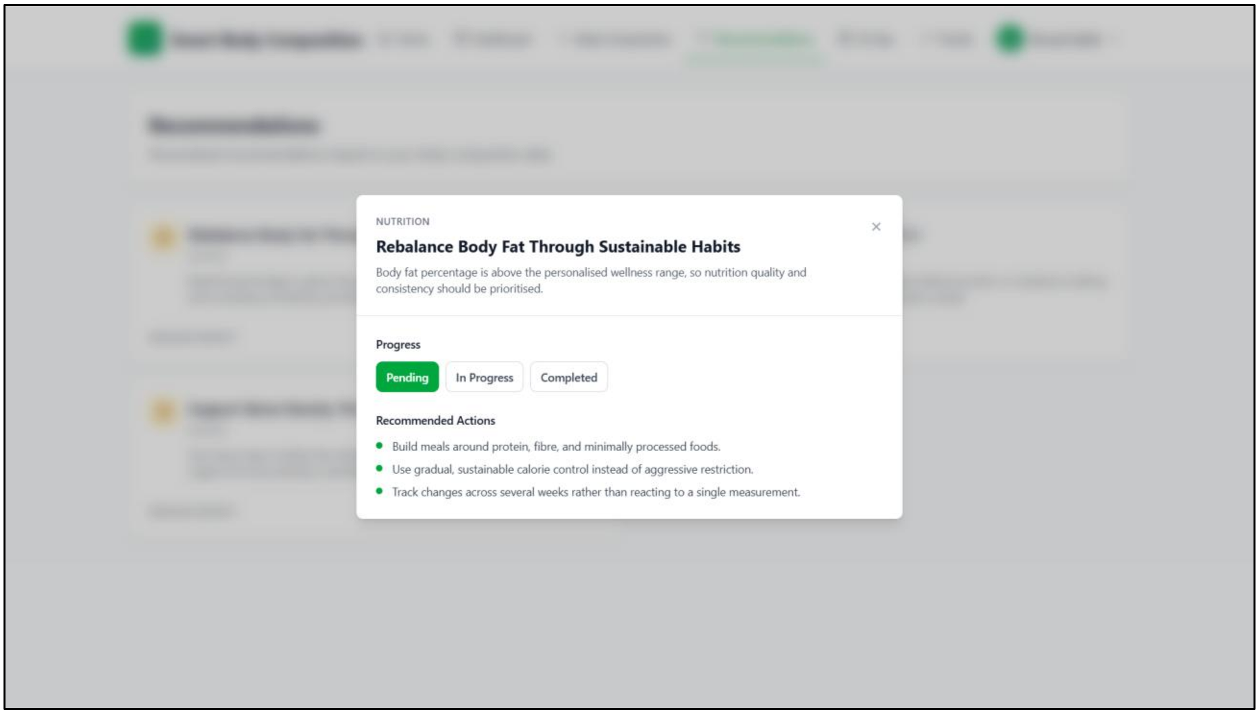
Understanding Recommendation Cards

Each card is structured to give you both context and actionable direction at a glance:

Element	Description
Title	The metric being addressed (e.g., "Body Fat Percentage")
Recommendation Text	Actionable advice tailored to your situation

Marking Recommendations

Once you have read a recommendation and decided how to act on it, you can mark it as read by clicking the checkmark or toggle on the card. A filter at the top of the page lets you switch between viewing all recommendations, only active ones, or only those you have already read.

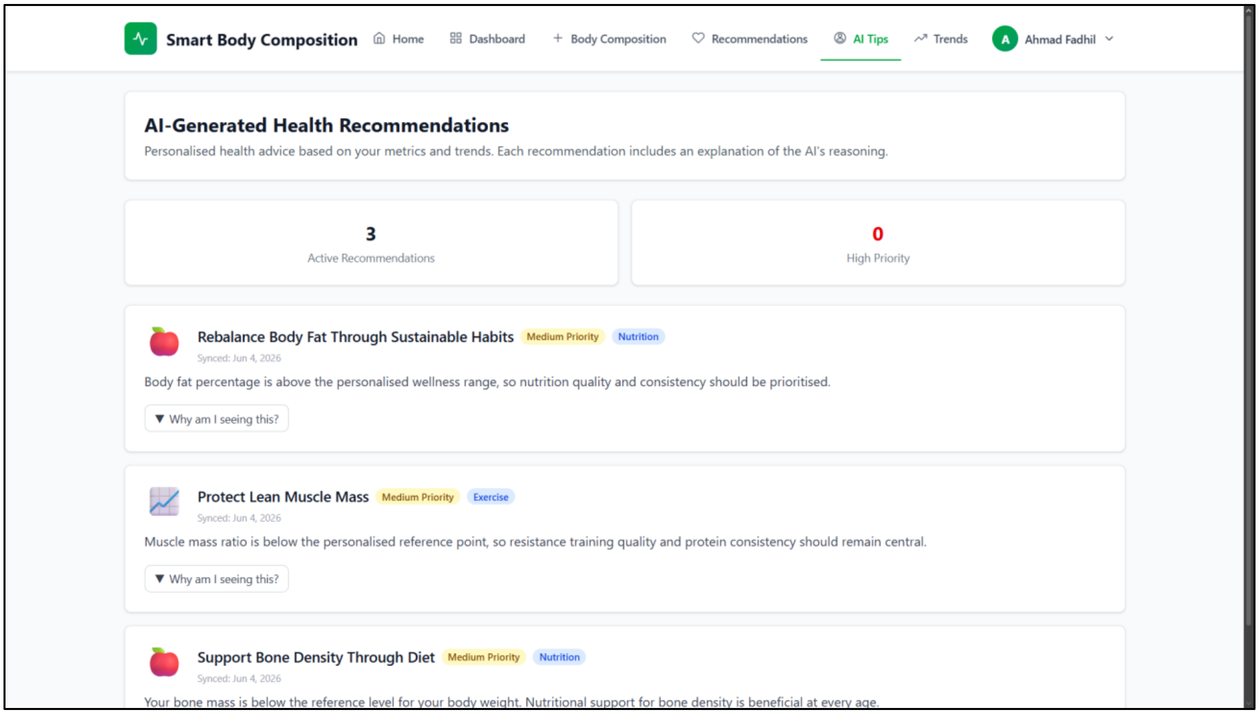


10. AI Tips

Where the Recommendations page delivers personalized insights drawn from your measurement history, the AI Tips feature offers something a little different: quick, general health guidance tailored to your profile — your age, gender, height, and overall trends. Think of it as a lighter, faster alternative when you want a nudge in the right direction without a full recommendation cycle.

Accessing AI Tips

34. Click AI Tips in the sidebar.



35. Tips are displayed as cards with brief, actionable suggestions.
36. The content covers general wellness, hydration, physical activity, and nutrition — areas that apply broadly rather than to a specific metric anomaly.

> **Note:** AI Tips are broad suggestions based on your profile. For deeper, measurement-based insights, use the Recommendations page — it looks at your actual trend data rather than general health principles.

11. Notifications

Consistency is the hardest part of any health tracking habit. The notification system is designed to help with that — keeping you informed about your progress and gently reminding you when it is time to log another measurement.

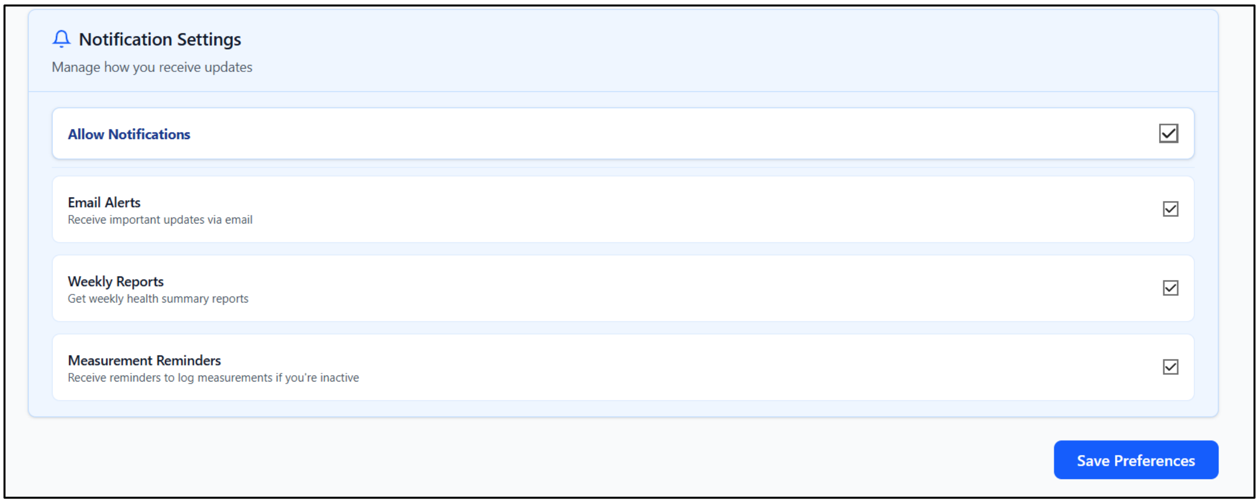
Types of Notifications

Type	Trigger	Content
Weekly Report	Automatically generated every week	Summary of your measurements and trends over the past 7 days
Measurement Reminder	Scheduled daily at 8:00 PM	Reminder to take and log your daily measurement
Goal Achieved	When you reach a goal target	Congratulations and summary of your achievement

Recommendation Generated	After generating new recommendations	Alert that new health insights are available

Notification Preferences

If you find certain notifications unnecessary, you can dial them back in Settings under the notification preferences section:



Toggle	What it controls
Notifications Enabled	Master switch for all notifications
Email Alerts	Receive notifications by email (requires email configuration by admin)
Weekly Reports	Enable/disable weekly summary reports
Measurement Reminders	Enable/disable daily measurement reminders

12. Admin Panel

The admin panel is the control room for the entire system, and access is restricted to users with the admin role. From here, an administrator can manage user accounts, review all data in the system, edit the recommendation templates that drive the advice engine, and configure system-wide settings like session timeouts and mail credentials.

Accessing the Admin Panel

Admin accounts see an additional Admin menu group in the sidebar that is invisible to regular users. If you are evaluating the system and need to explore the admin features, use the demo credentials below:

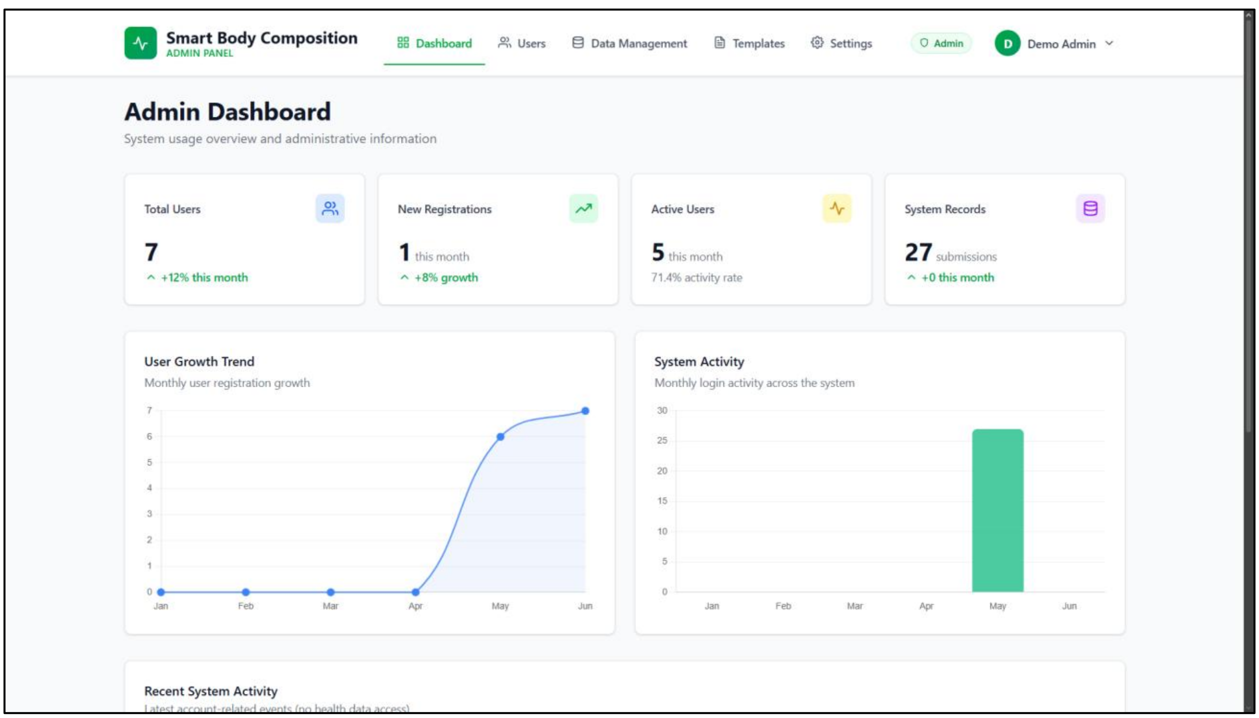
Email	Password
admin.demo@example.com	password

Admin Features

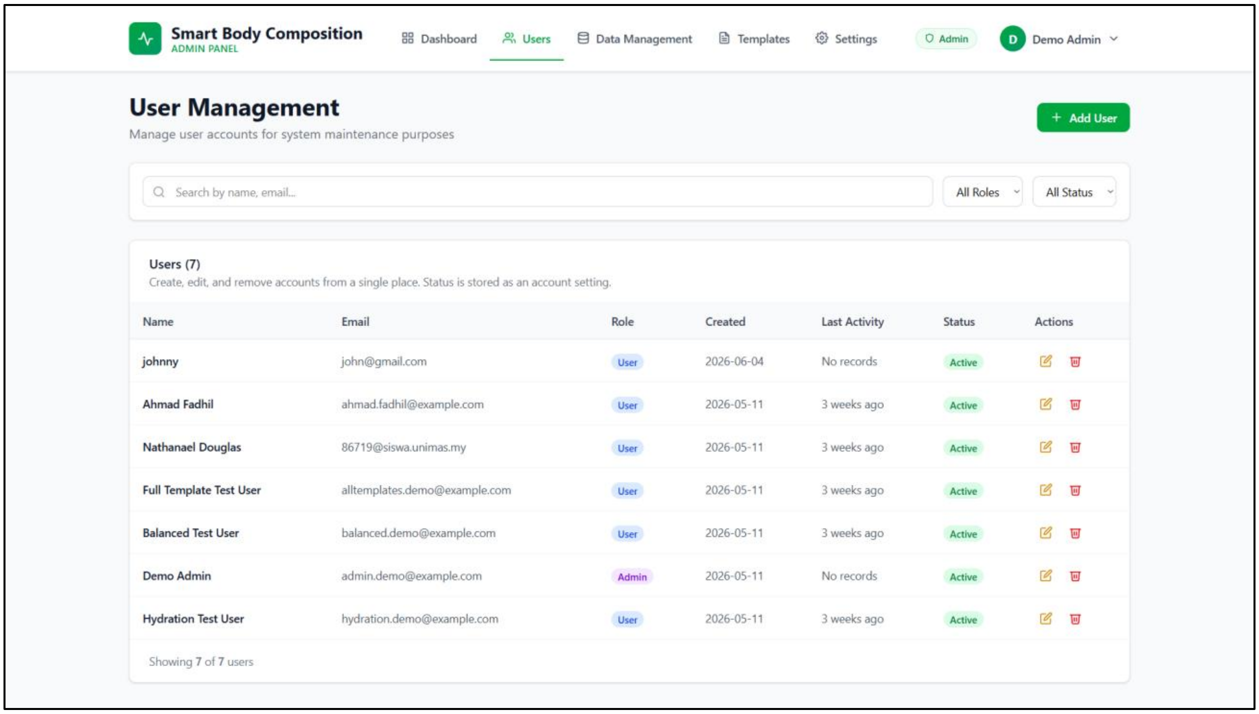
Dashboard

The admin dashboard gives a high-level health check of the entire system — not just one user's data, but everything:

- Total users — number of registered accounts.
- Total records — how many body composition entries exist across all users.
- Recent activity — the latest registrations and measurements logged by any user.
- System health — quick status indicators.

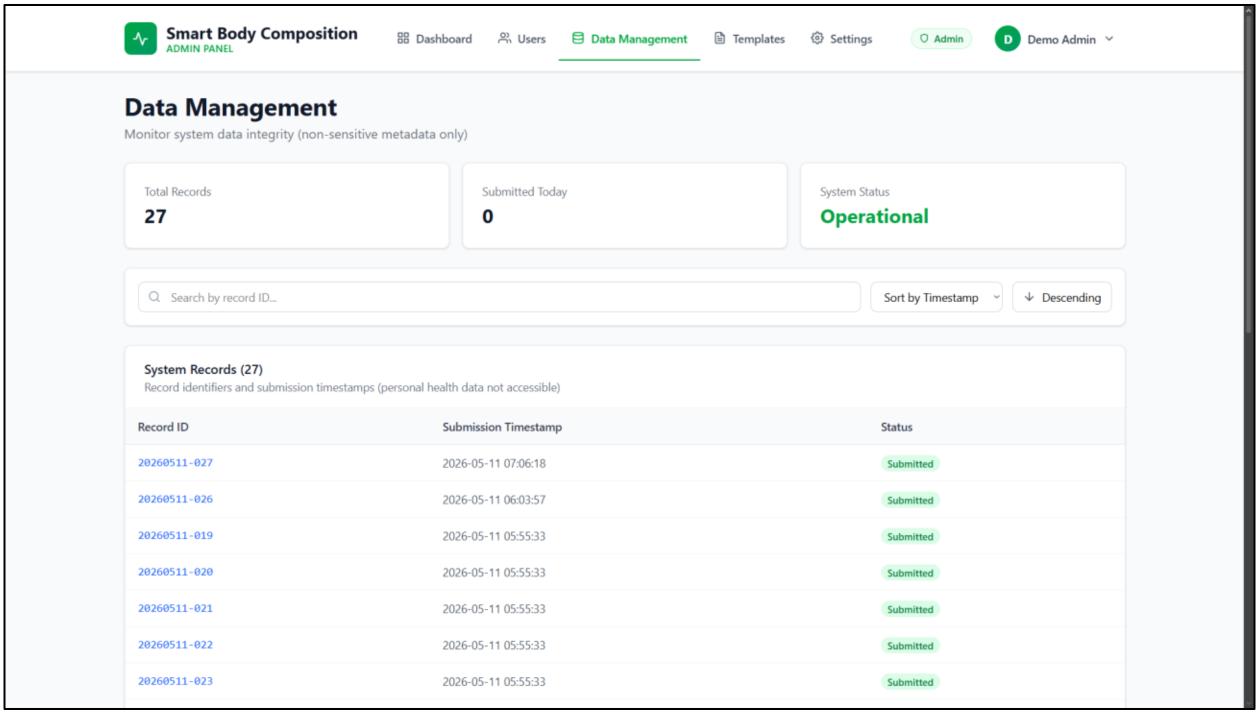


Manage Users



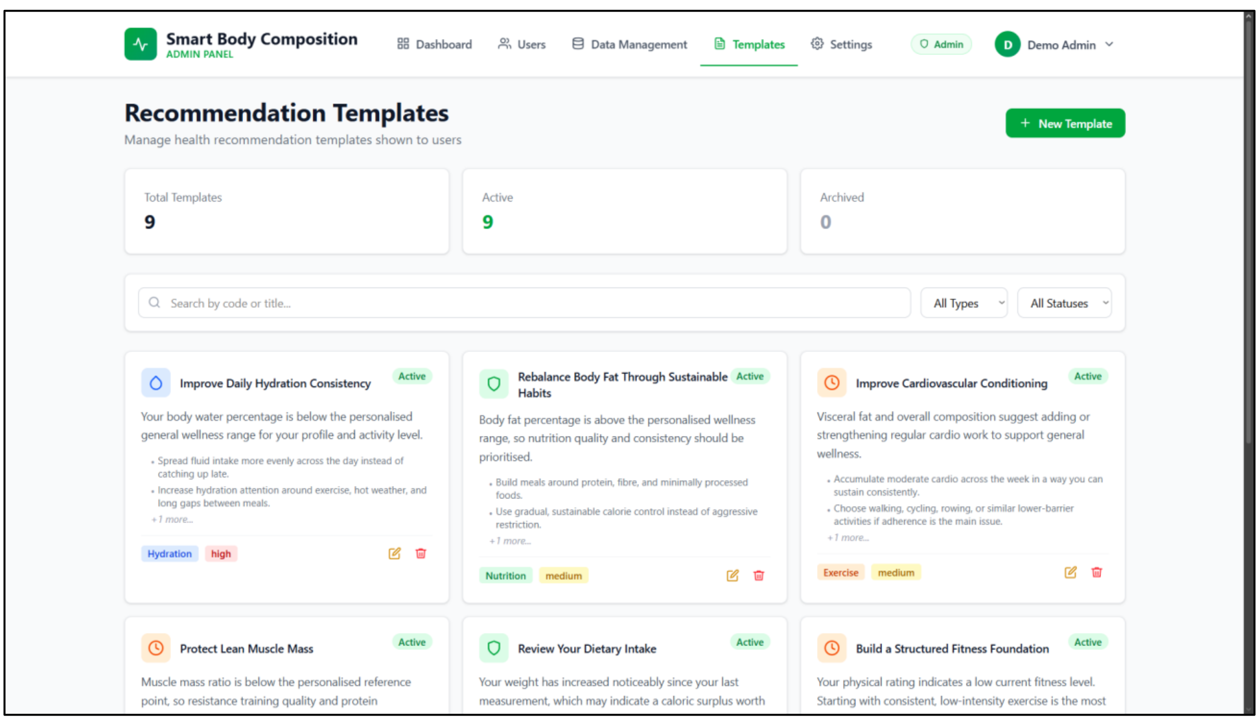
The user management page is where most day-to-day administration happens. You can search the full user list, create new accounts manually, edit any user's profile details (name, email, age, gender, height), toggle account status to enable or disable access, and view the body composition history of any individual user.

View Records



A system-wide view of all body composition records. Useful for auditing data quality or tracking system-wide usage patterns.

Recommendation Templates

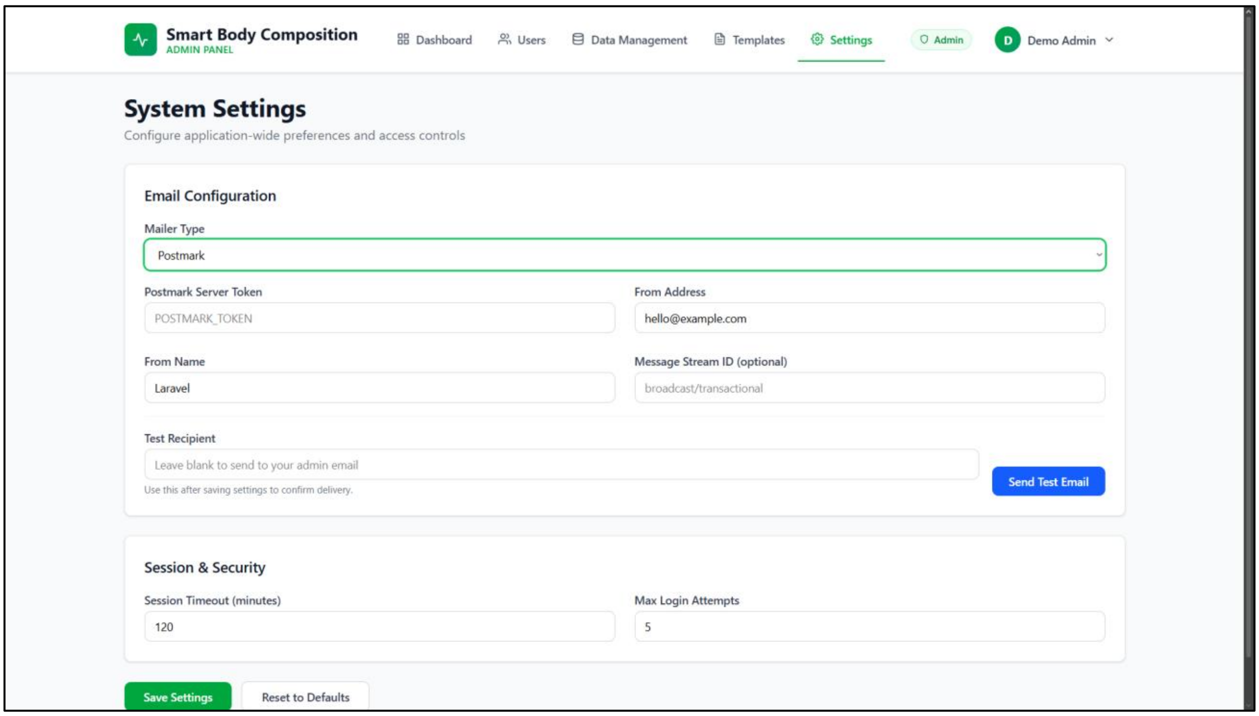


This is where the intelligence behind the recommendation engine lives. Each template defines a rule: if a user's metric falls into a particular range or shows a particular trend, serve this advice card with this severity level. Administrators can create new templates, edit the advice text and thresholds on existing ones, and toggle templates between active and archived states.

Each template contains the following fields:

- Code — a unique identifier (e.g., "HIGH_BODY_FAT").
- Type — the metric category (body_fat, muscle_mass, visceral_fat, hydration, etc.).
- Title — the short heading shown on the recommendation card.
- Summary — a single-line overview of the recommendation.
- Details — the full advice text in JSON format, which supports multiple paragraphs.
- Priority — the severity level assigned to this recommendation (Low / Moderate / High).
- Status — whether the template is Active or Archived.

System Settings



System-wide behaviour is configured here. The most relevant settings for a new deployment are:

- Session Timeout — how long the system waits before logging out an inactive user (default: 120 minutes).
- Max Login Attempts — the number of failed login attempts allowed before a lockout is triggered (default: 5).
- Email Configuration — SMTP server settings needed for the system to send notification emails.
- Test Email — a utility for verifying that the mail configuration is working correctly.

⚠ **Important:** Email notifications will only work once the SMTP mail settings are properly configured. Out of the box, the system uses MAIL_MAILER=log, meaning emails are written to the application log file rather than being sent — which is fine for local development but needs to be changed for a live deployment.

13. Troubleshooting

If something is not working as expected, this section is your first port of call. The issues listed here account for the vast majority of problems reported during development and testing. Work through the relevant solutions in order — most problems are resolved by the first or second step.

"Database connection refused" error

Symptoms: This error appears when running php artisan migrate or when trying to open the application in a browser.

Solutions:

37. Make sure MySQL is running in the XAMPP Control Panel — both Apache and MySQL should show a green "Running" status.
38. Double-check the database settings in your .env file:

```
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USERNAME=root
DB_PASSWORD=              (empty for default XAMPP)
```

39. Confirm that the smartbodycomposition database has been created in phpMyAdmin at http://localhost/phpmyadmin.

"Vite manifest not found" or blank white page

Symptoms: The browser displays a completely blank page, or the application log contains a "Vite manifest not found" error.

The frontend assets have not been compiled yet — or need to be recompiled after a code change. Run:

```
npm run build
```

If you are actively developing and want the browser to reflect changes in real time without running a full build each time, use:

```
npm run dev
```

"Specified key was too long" error

Symptoms: The migration fails with an error message about key length.

This is caused by older versions of MySQL (below 5.7.7) that have smaller index key size limits. Add the following to app/Providers/AppServiceProvider.php inside the boot() method:

```
use Illuminate\Support\Facades\Schema;

public function boot(): void
{
    Schema::defaultStringLength(191);
}
```

Port 8000 already in use

Symptoms: php artisan serve fails with "Failed to listen on 127.0.0.1:8000".

Another process on your machine is already using port 8000. Start the server on a different port instead:

```
php artisan serve --port=8080
```

Then access the application at http://127.0.0.1:8080.

Application shows "500 Server Error"

Symptoms: A generic 500 error page appears instead of the application.

The most reliable way to diagnose a 500 error is to inspect the Laravel log file directly:

```
# Windows (PowerShell)
Get-Content storage\logs\laravel.log -Tail 50

# Linux / macOS
tail -50 storage/logs/laravel.log
```

The most common root causes are: a missing .env file (fix with copy .env.example .env), a missing APP_KEY (fix with php artisan key:generate), incorrect database credentials in .env, or missing PHP extensions. You can check which extensions are loaded by running php -m and confirming that pdo_mysql, mbstring, zip, and gd are all present.

Cannot log in after registration

40. Ensure the database has been seeded: php artisan migrate --seed
41. Clear the application cache: php artisan cache:clear
42. Check storage/logs/laravel.log for any authentication-related errors.
43. If using the demo accounts provided in this manual, make sure the email address and password are entered exactly as shown — they are case-sensitive.

npm install fails with errors

44. Verify your Node.js version is 20.x or higher: node --version
45. If the version is correct but errors persist, delete the existing node_modules directory and lock file, then re-run the install:

```
Remove-Item -Recurse -Force node_modules   # Windows
Remove-Item package-lock.json
npm install
```

46. If issues continue, try clearing npm's local cache first: npm cache clean --force

Composer install takes too long or fails

47. Check your internet connection — Composer downloads packages from external repositories and will fail or stall if connectivity is intermittent.
48. If the download times out, increase Composer's timeout limit: composer config process-timeout 2000
49. To see exactly where Composer is getting stuck, run with verbose output: composer install -vvv

Apache won't start in XAMPP

50. Check whether another application is already using port 80 (common culprits include Skype, IIS, and VMware):

```
netstat -ano | findstr :80
```

51. If port 80 is occupied, change Apache's listening port in XAMPP Control Panel → Config → Apache (httpd.conf) by updating Listen 80 to Listen 8080.

"class not found" or "permission denied" errors

52. Regenerate Composer's autoload map:

```
composer dump-autoload
```

53. Clear Laravel's cached configuration, cached application data, and compiled views in one pass:

```
php artisan config:clear
php artisan cache:clear
php artisan view:clear
```

— End of User Manual —

For further assistance, please open an issue on the GitHub repository (https://github.com/natd23111/Smart-Body-Composition-System) or contact the project author directly.